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KEY CONCEPTS IN MACROECONOMICS

WELCOME TO THE PRESENTATION

Submitted by

- H M Yeasin

ID: 20245103021

- Md. Fahim Islam

ID: 20245103035

- Md Sakibur Rahman

ID: 20245103091

- Abu Ahmad Mahfuzor Rahman

ID:20245103092

- MD.Abdur Rahim

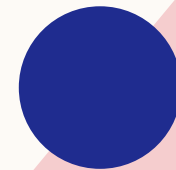
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Submitted to

Mst. Nazmin Akter

Lecturer

Department of Economics



TOPIC COVERED

1. COMPONENTS OF GDP
2. GDP COMPONENTS – REAL-WORLD EXAMPLES
3. WHY BANKS DON'T HOLD 100% RESERVES
4. HOW MONEY IS CREATED – THE MULTIPLIER EFFECT
5. INFLATION VS. UNEMPLOYMENT – SHORT & LONG RUN



COMPONENT OF GDP

- **Gross Domestic Product (GDP)** measures total economic output.

- **Four Components:**

1. **Consumption (C)** – Household spending (e.g., buying food, clothes)

2. **Investment (I)** – Business expenditures (e.g., purchasing machinery)

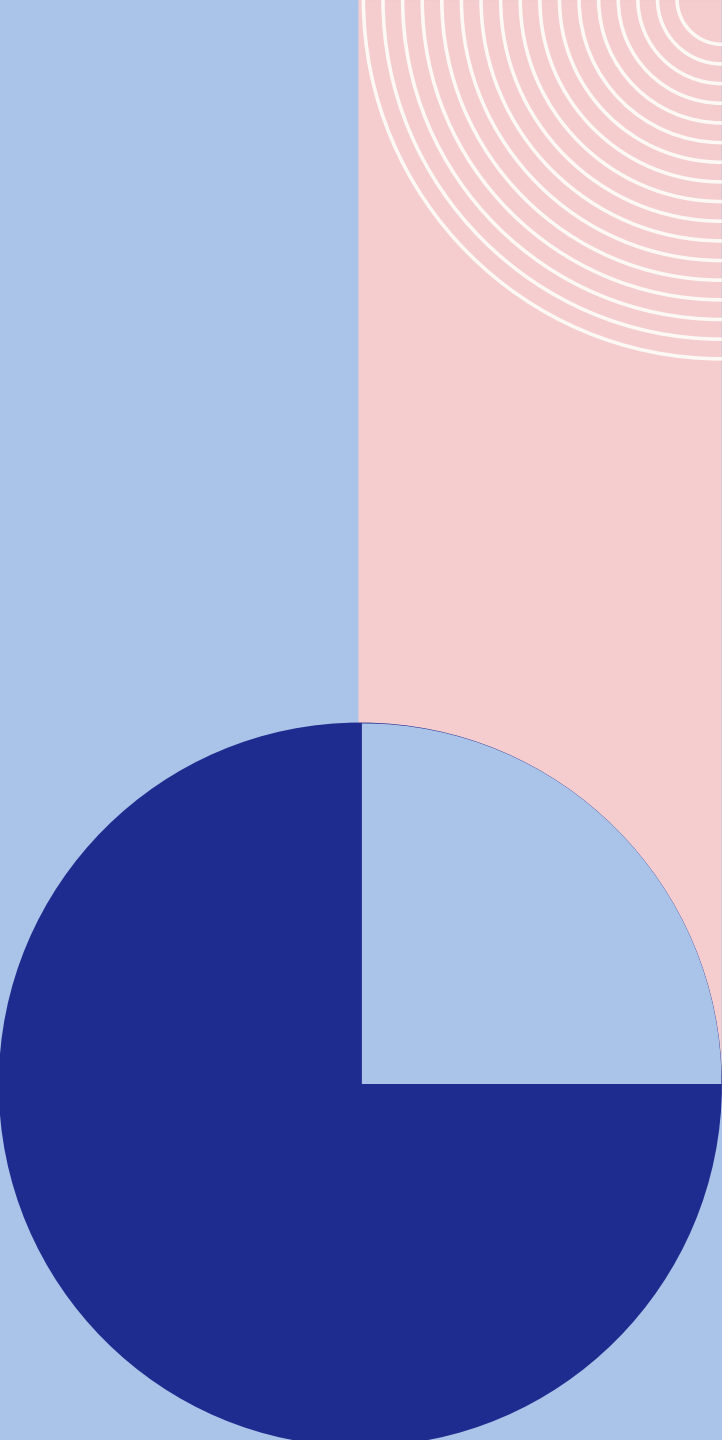
3. **Government Spending (G)** – Public services and infrastructure (e.g., education, roads)

4. **Net Exports (NX = Exports – Imports)** – International trade (e.g., exporting electronics)



GDP COMPONENTS – REAL-WORLD EXAMPLES

- **Consumption:** Family buys a new car
- **Investment:** Company builds a new factory
- **Government Spending:** State funds public school renovations
- **Net Exports:** Country exports \$2B in cars, imports \$1.5B in oil → $NX = \$0.5B$



WHY BANKS DON'T HOLD 100 PERCENT RESERVES

- Profit Motive:**

Banks earn money mainly by lending out a portion of the deposits they receive and charging interest on those loans. Holding 100% of deposits as reserves would mean they can't lend and therefore can't earn interest income.

- Efficient Use of Funds:**

Not all depositors withdraw their money at the same time. By keeping only a fraction of deposits as reserves (typically 10% or less), banks can use the rest to support loans and investments.

- Monetary Policy and Economic Growth:**

Fractional reserve banking supports credit expansion, which stimulates economic activity. Central banks regulate this process to manage inflation and economic growth.

MONEY CREATION PROCESS IN FRACTIONAL RESERVE BANKING

- Initial Deposit:

Suppose someone deposits \$1,000 in a bank. If the reserve requirement is 10%, the bank must hold \$100 and can lend out \$900.

- Loan and Redeployment:

The \$900 is lent to a borrower, who then spends it. The recipient of that money deposits it into another bank. That bank now has a \$900 deposit. It keeps \$90 (10%) as reserves and lends out \$810.

- Cycle Repeats:

This process repeats across the banking system. With each round, a portion of the money is kept as reserves and the rest is lent out again.



- **Money Multiplier Effect:**

The total money created in the system is greater than the initial deposit. The theoretical money multiplier is calculated as:

$$\text{Money Multiplier} = 1 / \text{Reserve Ratio} = 1 / 0.10 = 10$$

So the initial \$1,000 deposit can ultimately lead to up to \$10,000 in total money supply (under ideal conditions with full re-lending and redepositing).

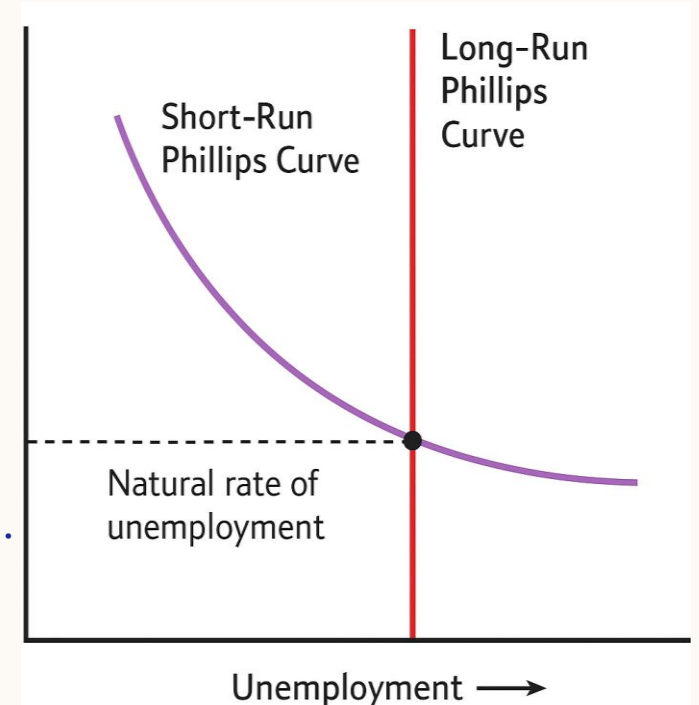
INFLATION AND UNEMPLOYMENT: SHORT-RUN VS LONG-RUN TRADE-OFF

1. Short-Run Trade-Off

- **Inverse relationship** between inflation and unemployment.
- $\uparrow$ Inflation $\rightarrow$ $\downarrow$ Unemployment (due to increased demand).
- Caused by **sticky wages/prices**.
- Illustrated by the **downward-sloping SR Phillips Curve**.

Long-Run Trade-Off

- No trade-off: **Unemployment returns to its natural rate (NAIRU)**.
- **Expectations adjust**, wages rise with inflation.
- **Long-Run Phillips Curve (LRPC)** is vertical.



Example:

- Unemployment falls from **8% to 4%**.
- Inflation rises from **1% to 5%**.

Time Frame	Trade Off	Phillips Curve Shape
Short Run	Yes(Inverse)	Downward Sloping
Long Run	No	Vertical at Natural Rate



CONCLUSION

- **GDP** is made up of consumption, investment, government spending, and net exports.
- **Banks use fractional reserves** to lend money and expand the money supply.
- **Short-run trade-off** exists between inflation and unemployment, but in the **long run**, this trade-off disappears.

THANK YOU

